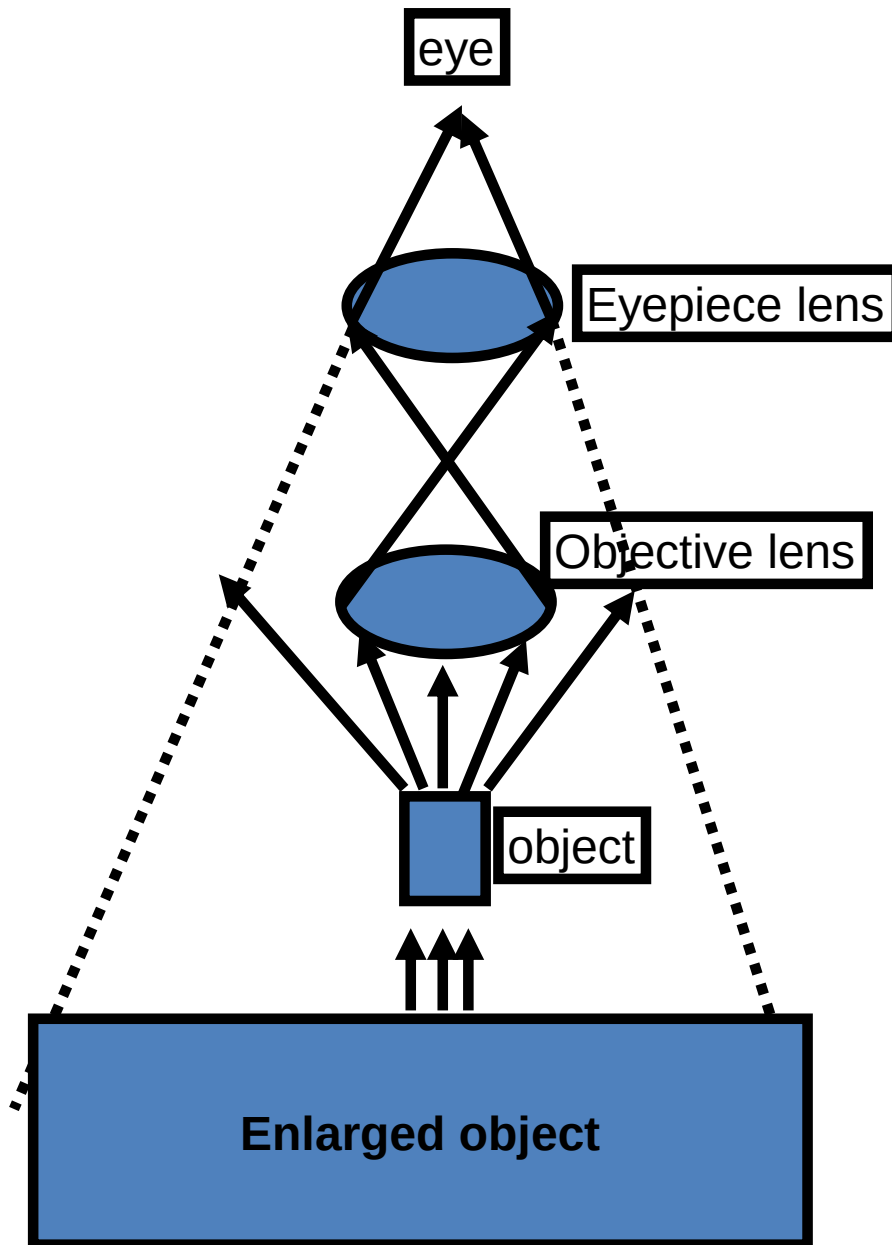


Optical microscopy



X-ray microscope

- The scattered X-rays cannot, however, normally be focused by any currently known experimental technique because no convenient material that can focus X-rays (in the way that glass lenses focus visible light) has yet been devised.

Thus, X-ray microscope does not exist.

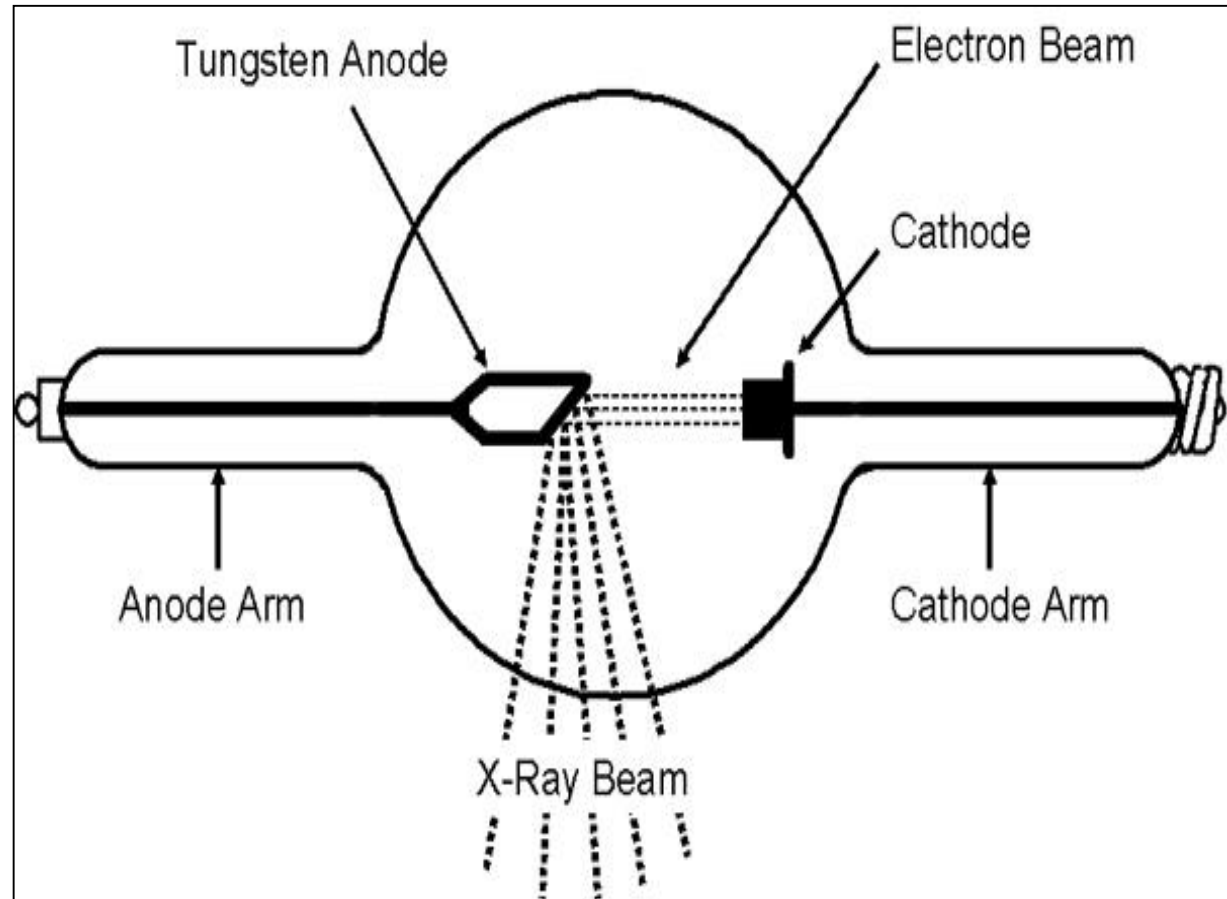
- Various mathematical calculations act as lens in X-ray diffraction thereby enabling us to 'see' atoms in a molecule.
- Solving '**phase problem**' by various mathematical calculations is like introducing '**lens**' in X-ray diffraction.

Sealed tube

- Suitable for small molecule

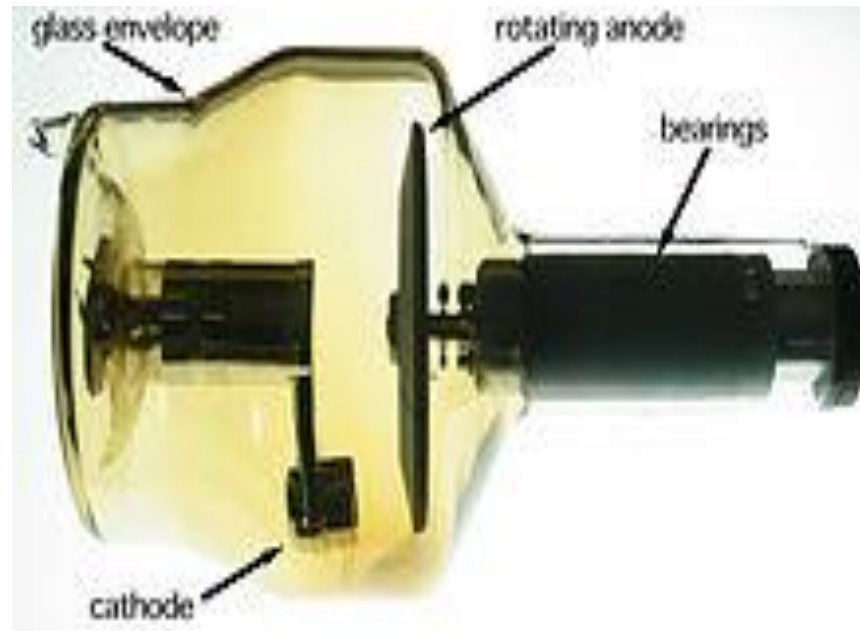


X-ray tube

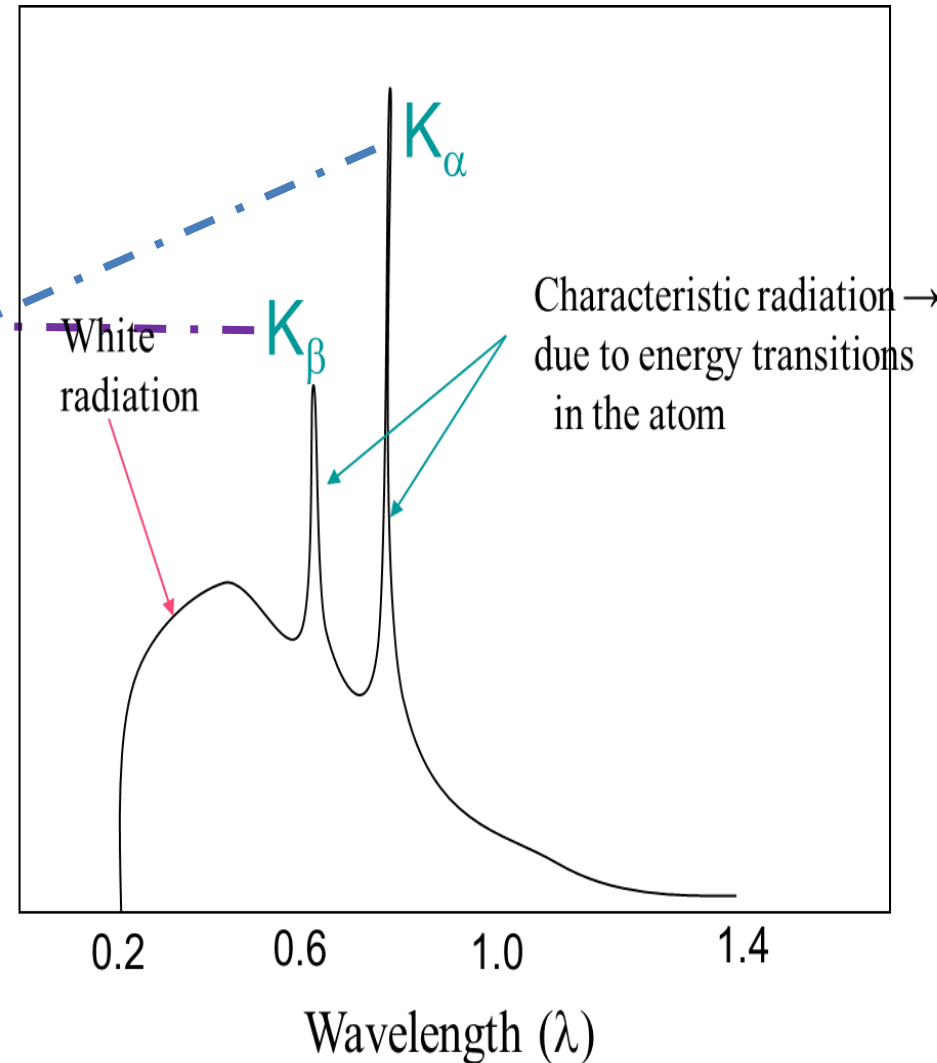
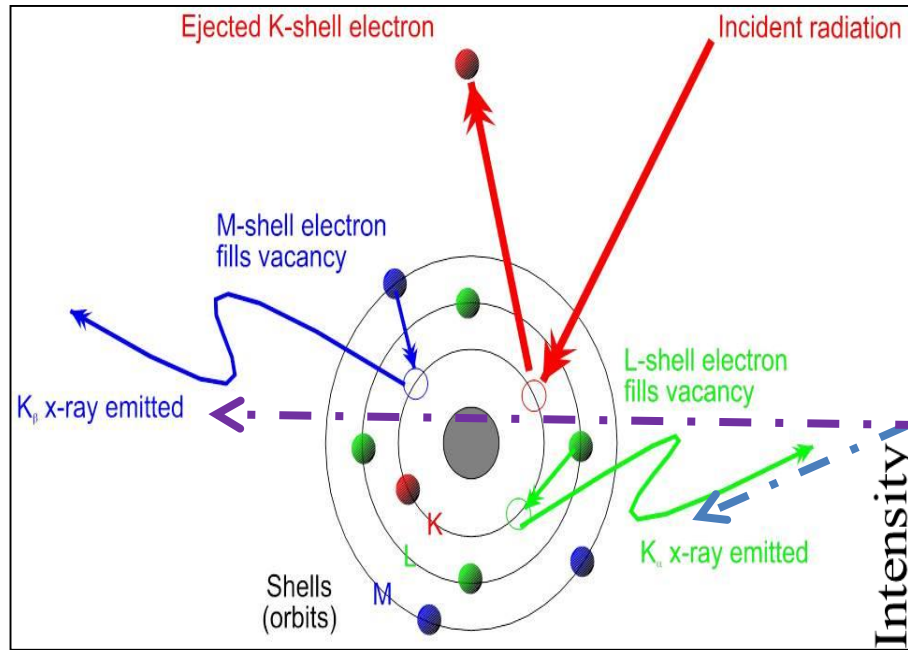


Rotating anode

- High flux X-ray, suitable for macromolecules such as protein crystals



How X-rays are generated



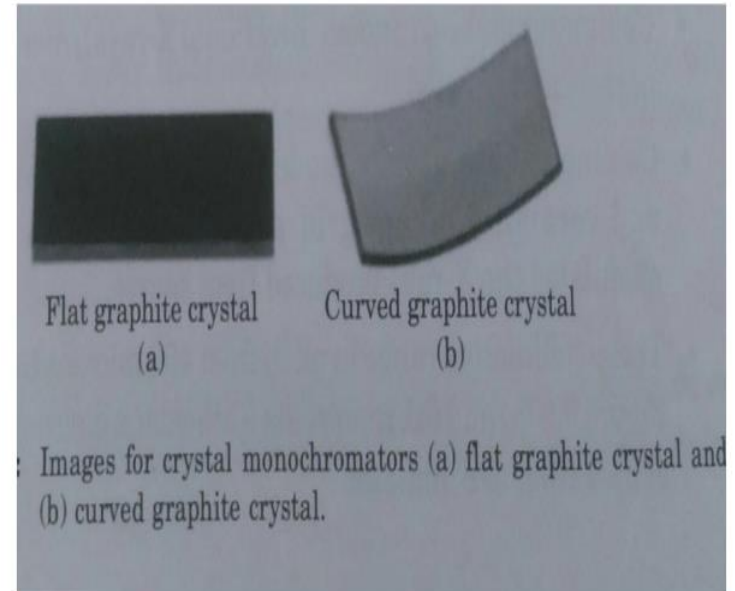
Target Metal	λ Of K _α radiation (Å)
Mo	0.71
Cu	1.54
Co	1.79
Fe	1.94
Cr	2.29

Various X-rays and filters, Z-1 rule

22 Ti Titanium 47.887	23 V Vanadium 50.9415	24 Cr Chromium 51.9961	25 Mn Manganese 54.938045	26 Fe Iron 55.845	27 Co Cobalt 58.933195	28 Ni Nickel 58.6934	29 Cu Copper 63.546	30 Zn Zinc 65.38
40 Zr Zirconium 91.224	41 Nb Niobium 92.90638	42 Mo Molybdenum 95.94	43 Tc Technetium (97.9072)	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.90550	46 Pd Palladium 106.42	47 Ag Silver 107.8682	48 Cd Cadmium 112.411

- Flat crystal Graphite monochromator
- Bent crystal Graphite monochromator
- Multilayer optics

Elements	(KV)	λ Of $K_{\alpha 1}$ radiation (Å)	λ Of $K_{\alpha 2}$ radiation (Å)	λ Of K_{β} radiation (Å)	K_{β} -Filter (mm)
Ag	25.52	0.55941	0.5638	0.49707	Pd 0.0461
Mo	20	0.7093	0.71359	0.63229	Zr 0.0678
Cu	8.98	1.540598	1.54439	1.39222	Ni 0.017
Ni	8.33	1.65791	1.66175	1.50014	Co 0.0158
Co	7.71	1.78897	1.79285	1.62079	Fe 0.0166
Fe	7.11	1.93604	1.93998	1.75661	Mn 0.0168
Cr	5.99	2.2897	2.29361	2.08487	V 0.169



Images for crystal monochromators (a) flat graphite crystal and (b) curved graphite crystal.

